

Project Initialization and Planning Phase

Date	29/09/2026
Team ID	UNM1425comiccraft
Project Title	comiccraft
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to create personalized AI-generated comic stories and illustrations from user prompts. It combines FastAPI, Gemini Flash and Gemini Pro with Stable Diffusion to automate story outlines, narration, dialogues, panel images, web preview, and PDF export.

Project Overview	
Objective	To provide an end-to-end web application that converts user prompts into personalized comic stories, dialogues, illustrations, and a downloadable PDF.
Scope	The project covers prompt collection, 5-panel outline generation, narration and dialogue generation, image generation, panel layout, web preview, and PDF export using FastAPI and Jinja2.
Problem Statement	
Description	Creating comics manually can require separate skills for story writing, illustration, layout, and export. ComicCraft integrates these steps into one AI-assisted workflow.
Impact	The integrated workflow makes comic creation accessible to non-artists and casual users and provides a cohesive output that can be viewed, saved, printed, or shared as a PDF.
Proposed Solution	
Approach	Use Gemini Flash for structured 5-panel outlines, Gemini Pro for narration and dialogue, Stable Diffusion for panel illustrations, and FastAPI/Jinja2 to coordinate input, processing, preview, and export.
Key Features	5-panel story outline and detailed narration/dialogue. - AI-generated comic illustrations with customizable story inputs.

	• Panel-by-panel web preview and Jinja2-based interface. • Downloadable PDF export with images and narration.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	GPU/compute resources for Stable Diffusion image generation	Exact hardware specification not stated
Memory	RAM / memory	Not specified
Storage	Disk space for application files, models and exports	Not specified
Software		
Frameworks	Python web framework	FastAPI
Libraries	AI, image, PDF and web libraries	google-generativeai, diffusers, transformers, fpdf, Pillow, accelerate, Jinja2, python-multipart, PyTorch
Development Environment	Python development tools	pip, virtual environment, Uvicorn; IDE not specified
Data		
Data	User inputs and generated comic content	Size and file format not specified